

1. Solution of Schrodinger's equation for the hydrogen atom results in quantum numbers that define the position and energy of electrons around the hydrogen nucleus.
 - a. List the 3 quantum numbers and for each, indicate the property of the electron that they define. [3 Marks]
 - b. State the key difference between solutions for the hydrogen atom and the helium atom. State the main reason for the difference. [2 Marks]
 - c. By considering two hydrogen atoms brought very close together, describe the changes in atomic orbitals during covalent bonding. [3 Marks]
2. Optics and Optical Fibres
 - a. Describe:
 - i. Three sources of attenuation in optical fibres. [4.5 Marks]
 - ii. The effect of dispersion in optical fibres on digital communication signals. [2.5 Marks]
 - b. An optical fibre link of length 10 km has attenuation of 0.25 dB/km. If input power is 1 mW, calculate the output power in mW. [3 Marks]
 - c. Explain why light can be coupled from an LED on a multi-mode optical fibre while coupling to a single mode optical fibre requires a laser. [3 Marks]
 - d. A step-index fibre has $n_1 = 1.46$ and $n_2 = 1.45$. Determine the numerical aperture and acceptance angle of the fibre. [5 Marks]
3. Optical resonators are key components of a laser.
 - a. Explain the purpose of an optical resonator in a laser. [3 Marks]
 - b. List the mechanisms that result in loss of light intensity in an optical resonator. [4 Marks]
 - c. Distinguish between a stable resonator and an unstable resonator. Indicate the condition for stability. [5 Marks]
 - d. Figure Q3 shows some common mirror arrangements in resonators. Determine the stability of the following resonator configurations: [3 Marks]
 - i. Spherical
 - ii. Hemispherical
4. Figure Q4 shows the energy band diagrams for the active ions in the alexandrite and titanium-sapphire lasers. In both cases, the pumping and the lasing energy levels have spread into an energy band.
 - a. For each laser, state whether it is 3-level or 4-level. [2 Marks]
 - b. With reference to atomic energy level interactions and quantum mechanical principles, explain why some of the energy levels have spread into an energy band. [4 Marks]
 - c. Explain why solid-state lasers in which the active ions are transition metal ions are more likely to be tunable than those using lanthanide metal ions. (HINT: state the electronic structures of argon and xenon and compare them with a 3^+ ion). [4 Marks]

- d. The alexandrite laser can have the output wavelength tuned in the range 700–820 nm while the titanium-sapphire laser can have its output wavelength tuned in the range 650–1100 nm.
- Explain how the output frequency of a tunable laser can be adjusted. [2 Marks]
 - Determine the lowest and highest energy transition in the alexandrite laser in eV. [3 Marks]
 - Determine the lowest and highest energy transition in the titanium-sapphire laser in eV. [3 Marks]

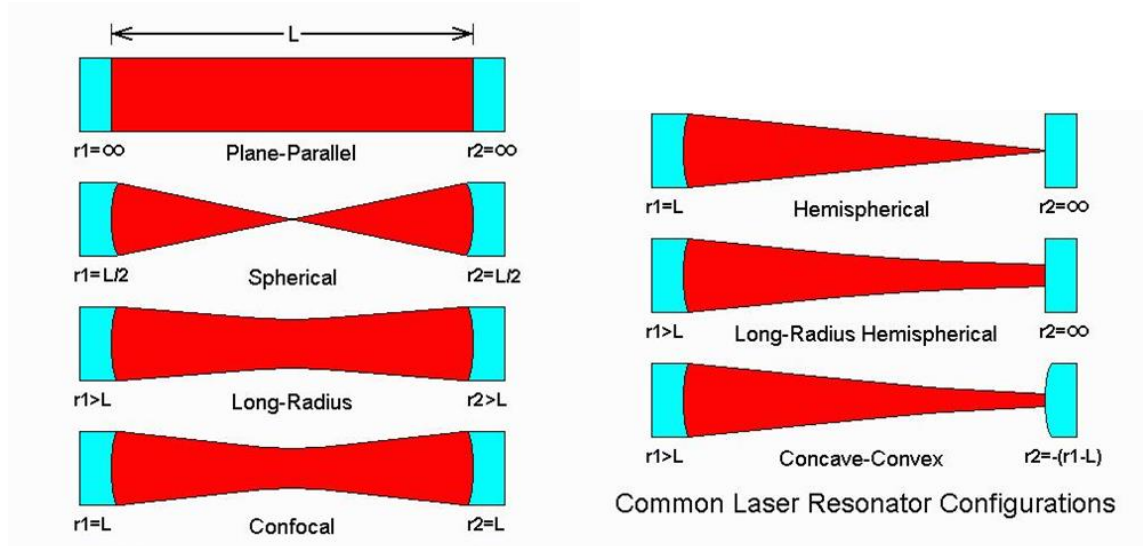


Figure Q3: Common Resonator Mirror Configurations

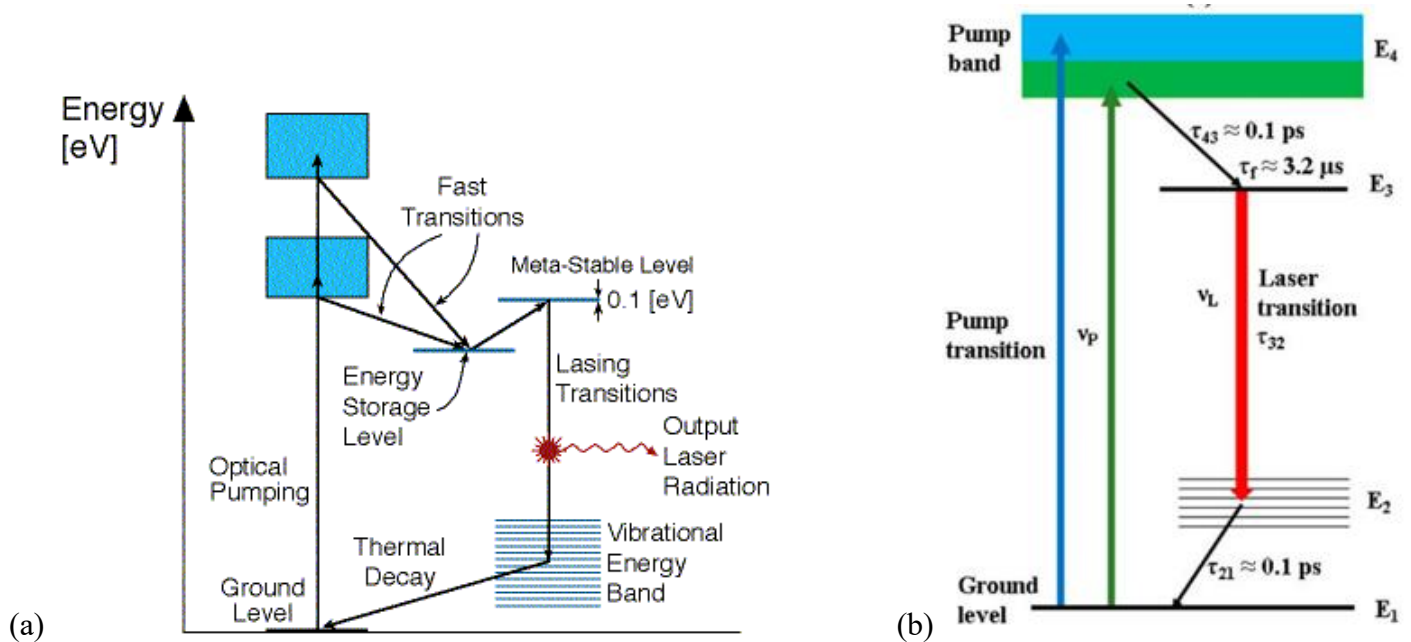


Figure Q4: Alexandrite and Titanium-Sapphire Laser Transitions